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Resource Group Scope Pattern

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Overview

The Resource Group Scope Pattern is the simplest deployment pattern in the Release Engine framework. It demonstrates how to deploy a single Azure resource (storage account) within a resource group using Azure Verified Modules (AVM) with minimal configuration.

Architecture

This pattern deploys a single Azure Storage Account within an existing Resource Group.

```
graph TB
    subgraph RG ["Resource Group"]
        SA["Azure Storage Account<br/>(Standard_LRS default)"]
    end

    style RG fill:#e1f5fe
```

style SA fill:#b3e5fc

Prerequisites

- **Azure Permissions:** Contributor access to target Resource Group
- **Service Principal:** Configured with appropriate permissions for storage account creation
- **Resource Group:** Must exist before deployment (not created by this pattern)

Resources Created

- **Azure Storage Account** (AVM: avm/res/storage/storage-account:0.27.1)
 - Uses Azure Verified Modules for best practices and security
 - Configurable location and name
 - Default SKU can be overridden via parameters

Parameters

Parameter	Type	Description	Default	Required
resourceLocation	string	Azure region for deployment	None	Yes
storageAccountName	string	Name of the storage account	None	Yes

Parameter Constraints

- **resourceLocation:** Must be one of: westeurope, uksouth
- **storageAccountName:** Must follow Azure storage account naming rules (3-24 characters, lowercase, numbers only)

Outputs

Output	Type	Description
storageAccountId	string	Resource ID of the created storage account

Deployment Scope

- **Target Scope:** Resource Group
- **Service Connection Required:** Yes
- **Multi-Stage:** No (single deployment stage)

Pipeline Configuration

Stage Configuration

```
stages:
- infrastructure:
  iac:
    name: single_resource_deployment
    deploymentScope: ResourceGroup
    serviceConnection: $(serviceConnection)
```

```

iacMainFileName: resource_group_scope_pattern.bicep
iacParameterFileName: ${
parameters.deploymentSettings.iacParameterFileName }}

```

Environment Variables Required

The pattern expects these variables to be defined in your configuration repository:

```

# In vars-{environment}.yaml
variables:
  storageAccountSku: Standard_LRS # or Standard_GRS for production
  resourceLocation: westeurope    # or your preferred region

```

Configuration Repository Integration

Parameter File Example

(resource_group_scope_pattern.parameters.json)

```

{
  "$schema": "https://schema.management.azure.com/schemas/2019-04-01/deploymentParameters.json#",
  "contentVersion": "1.0.0.0",
  "parameters": {
    "resourceLocation": {
      "value": "#{resourceLocation}#"
    },
    "storageAccountName": {
      "value": "st#{applicationAbbreviation}#{environmentAbbreviation}#{uniqueString}#"
    }
  }
}

```

Metadata Configuration

```

# In _config/metadata.yaml
variables:
  workload: "resource_group_scope_pattern" # Must match pattern
name
  applicationAbbreviation: "myapp" # Used in storage account
naming

```

Usage Examples

Development Environment

```

# vars-development.yaml
variables:
  environmentAbbreviation: dev
  resourceLocation: westeurope
  storageAccountSku: Standard_LRS # Cost-effective for development

```

Production Environment

```

# vars-production.yaml
variables:
  environmentAbbreviation: prd
  resourceLocation: westeurope
  storageAccountSku: Standard_GRS # Geo-redundant for production

```

Security Considerations

- **Azure Verified Modules:** Uses official Microsoft-maintained modules with security best practices
- **Managed Identity:** Storage account supports system-assigned managed identity (configurable via AVM parameters)
- **Encryption:** Encryption at rest enabled by default through AVM
- **Network Access:** Default configuration allows public access (can be restricted via AVM parameters)

Cost Optimization

- **Development:** Use Standard_LRS for lower costs
- **Production:** Consider Standard_GRS for geo-redundancy
- **Hot/Cool Tiers:** Configure access tiers based on usage patterns
- **Lifecycle Management:** Implement lifecycle policies for automatic tier transitions

Monitoring and Diagnostics

The storage account created by this pattern supports:

- **Azure Monitor Integration:** Metrics and logs automatically available
- **Diagnostic Settings:** Can be configured post-deployment
- **Application Insights:** Integration available for application-level monitoring
- **Cost Monitoring:** Built-in cost tracking and budgets

Troubleshooting

Common Issues

Storage Account Name Conflicts

Problem: Storage account name already exists globally **Solution:** Ensure storageAccountName includes unique suffixes like environment abbreviations

Permission Errors

Problem: Deployment fails with permission denied **Solution:** Verify service principal has Contributor role on target Resource Group

Region Availability

Problem: Deployment fails due to region restrictions **Solution:** Ensure selected region supports storage accounts and is in allowed list

Validation Commands

```
# Validate Bicep template
az bicep build --file resource_group_scope_pattern.bicep

# Test deployment (what-if)
az deployment group what-if \
  --resource-group myResourceGroup \
  --template-file resource_group_scope_pattern.bicep \
```

```
--parameters resourceLocation=westeurope  
storageAccountName=mystorageaccount
```

Pattern Evolution

This pattern serves as the foundation for more complex patterns. Consider graduating to:

- **Multi Resource Pattern:** When you need multiple related resources
- **Multi Stage Pattern:** When you need complex dependencies
- **Application Pattern:** When deploying full applications with storage dependencies

Maintenance Notes

- **AVM Updates:** Regularly update to latest AVM versions for security patches
- **Parameter Validation:** Add additional parameter validation as requirements evolve
- **Output Enhancement:** Add more outputs as downstream consumers require additional information

Related Patterns

- **Multi Stage Pattern:** Complex deployments with dependencies
- **Subscription Scope Pattern:** Subscription-level resource creation
- **Application Patterns:** Full application stacks including storage

This pattern demonstrates the simplest deployment scenario and serves as an excellent starting point for teams new to the Release Engine framework.